

AJAY VENKATESH

Brooklyn, NY

+1 516 585 9013 av3855@nyu.edu linkedin.com/in/ajay-venkatesh-32013b17b github.com/ajayv-19

Education

New York University | *Master of Science in Computer Engineering (GPA – 3.9/4)*

Aug 2024 – Aug 2026

Experience

Amazon

May 2025 – Aug 2025

Software Development Engineer Intern

Seattle, US

- Designed a secure UI Console architecture using Midway SSO, OAuth, and HELIS to enforce least-privilege, role-based access; presented the solution to the team and received design approval.
- Built serverless infrastructure using AWS CDK to deploy API Gateway, Lambda (Java 21), DynamoDB, and S3 with cross-account access; implemented CI/CD pipeline for multi-stage deployments.
- Developed the frontend using React 18, integrating AWS UI components, Midway AuthN, and HELIS AuthZ with role-aware views and filtering; delivered a UI that was 40% faster and significantly more secure than older version.
- Implemented backend APIs using Java Coral, Smithy models, Dagger DI, and CBOR serialization; improved response time by 35% and ensured reliability through comprehensive unit and integration testing.

New York University

Aug 2024 – May 2025

Software Developer (On-Campus)

New York, US

- Contributed to an AI-driven Remote Patient Monitoring System, awarded \$100K by the National Science Foundation, USA, employing React Native, TypeScript, and Cognito authentication for real-time health monitoring.
- Authored parallelized ETL pipelines in Python to convert unstructured NoSQL data into structured SQL using GCP Cloud Functions. Leveraged concurrent processing of data chunks, reducing transformation time by 60%.
- Developed a React website to host QuickSight dashboard handling 50K+ records, utilized Cognito, Amplify, and Lambda functions to establish login, role-based access control, and Stripe integration for payment gateway.

PricewaterhouseCoopers (PwC)

Aug 2021 – Aug 2024

Senior Software Engineer

- Developed a B2B responsive Power Pages (Dynamics 365 Portal) app for customer acquisition and onboarding, including importing external data and visualization using Power BI, resulting in a 40% reduction in onboarding time and a 30% increase in accuracy of customer journey tracking.
- Created and implemented a customer-facing portal with live chat functionality (Omnichannel) to automatically generate cases, contacts, and tasks, which directly reduced manual case entry by 70% and customer support response time by 25%.
- Developed custom configurations on both the client and server sides using JavaScript, plugins, workflows, and webhooks within Dynamics CRM and optimized server response time by 35%, lowering query execution time from 1.5s to less than 1s.
- Handled managed end-user accounts, permissions, and security roles on Dynamics 365, developing 6 specific security plans utilizing Azure security groups, which achieved 100% fewer incidents of unauthorized access and optimized account provisioning time by 50%.
- Created and implemented custom controls in Power Apps Component Framework (PCF) to design reusable UI elements with sophisticated data binding features and responsive design, which cut down on redundant UI development work by 60% and raised user satisfaction ratings by 15%.
- Developed Power Automate flows & Plugins to simplify the automation of business processes such as validation & notification, and reduced import time for 100k data to 1 hour compared to 5 hours (achieved 80% efficiency gain) and helped save \$15k each year.
- Combined 5 Power BI dashboard solutions into portals and applications to analyze data, resulting in a 60% acceleration of executive decision-making by shortening turnaround on reports from 2 days to real-time.
- Developed Low-Level Architecture design for MSDVI On Board, meeting technical specifications and adhering to SDLC guidelines, which led to a 25% decrease in defects during production and a \$10k decrease in rework expenses per release cycle.
- Engineered and optimized serverless database architecture (Dataverse/NoSQL) to improve data storage, access, and processing workflows, resulting in a 20% decrease in latency and 15% increase in transactions per second.
- Developed and customized Power Portals using Liquid and web templates, incorporating advanced form functionality and entity permissions, which cut manual configurations by 50% and optimized portal loading by 20%.

Projects

Advanced Expense Tracker (Java Desktop Application) | *Java, JavaFX, JDBC, SQLite, JFreeChart*

- * Architected a modular Java application supporting both individual and group expenses with secure login, leveraging OOP principles and design patterns (MVC and DAO) for separation of concerns.
- * Implemented concurrent transaction handling and real-time cost splitting using multithreading and synchronized blocks to ensure thread-safe database updates in group expense scenarios.

Skills

Programming Languages: C#, DAX, JavaScript, Typescript, SQL, Python

Web Development: HTML, CSS, PowerApps, Power Portals, React, Node.js, Express.js, Spring Boot

Database: Dataverse, RDBMS, NoSQL

Cloud: Azure (AI Services, Logic Apps, Databricks, Function Apps, Storage Blobs, Security Groups)

Tools: Git, Azure DevOps, Visual Studio, Dynamics CRM 365, ERP, XRM Toolbox, .NET, Power Automate, Power BI

Methodology: Agile, Waterfall

AI/ML: Machine Learning, Deep Learning, Big Data, MLOps, Principal Component Analysis

Certifications

Power Platform App Maker Associate	Power Platform Developer Associate	Power BI Data Analyst Associate
Power Platform Solution Architect Expert	Azure Data Scientist Associate	Azure Data Fundamentals
Platform Functional Consultant Associate		